

Note: The client code and the server code need to be in sync for the data delivered by our server to be interpreted and displayed correctly by our client. Currently, our unstated agreement is to serve a list of strings out of the `/counts` call from the server. By continuously testing the code during development, you can find out whether something you changed is breaking the client as soon as it happens. It is strongly recommended to use the 'Show -> Next to the Code' option to have the code running side by side with the coding window and keep the code running all the time. Options are shown in Figure 2.

To make PWA related auditing and to check *Chrome-> developer* tools, it may be easier to use *Show -> In a New Window* so that you have more real estate to see your app and the *dev tools* options side by side, without code panes taking up screen space.

The `getCounts` method

Now let us flesh out the `getCounts` method. As explained in the code sections below, we basically initiate a *GET* request using the Request library, process the response using Cheerio, read the title from the body of the response, do some string manipulation to reduce the string to '`<nCases> <nDeaths>`', split it, and then process and return the cases and deaths back to the client using the `response.json()` method.

1. We get a reference to both the request and response in the `getCounts` method:

```
// now make function to get new counts
function getCounts(req, resp) {
```

2. Use the `requests` library to place a GET request to our stats Web page:

```
request(
{
```

```
  method: "GET",
  url: "https://www.worldometers.
info/coronavirus/"
},
```

3. Log errors and use Cheerio to load the response body into variable `$`:

```
(err, res, body) => {
  if (err) return console.
error(err);
  let $ = cheerio.load(body);
```

4. Reduce '`Coronavirus Update (Live): <nCases> Cases and <nDeaths> Deaths from COVID-19 Virus Outbreak – Worldometer`' to '`<nCases> <nDeaths>`':

```
  let title = $("title");
  console.log(title.text());
  let coronaupdate = title.text();
  let mnstr = coronaupdate
    .replace("Coronavirus Update
(Live):", "")
    .replace(
      "Deaths from COVID-19 Wuhan
China Virus Outbreak - Worldometer",
      ""
    )
    .replace("Cases and", "");
```

5. Use the string `split()` to array and process the first and third splits.

Note: `split()` gives a second element for the space even if `<space>` is the separator.

```
const numbers = mnstr.trim().split("
");
const infections = numbers[0];
const deaths = numbers[2];
```

6. In a Web API we can return the values as JSON formatted elements; however, for simplicity, we simply return the cases and deaths in a list. `Respons.json()` handles the conversion.

```
oldcounts[0] = "cases " + infections;
```

```
oldcounts[1] = "deaths " + deaths;
resp.json(oldcounts);
```

Note: Cheerio allows many more options using selectors in a syntax like jQuery. Feel free to experiment with using better selectors and attributes that can avoid replacing multiple strings, as done in the above code.

Client side processing

Let us now shift our focus to the client side where we have simple `index.html` in which we process the list of items sent by the server. As explained in Figure 7, the client basically uses a simple unordered list `<ul>` which is populated with the coronavirus statistics. The code below instructs `index.html` to use the client-side `script.js`, about which we will learn more in the next section.

```
<!-- import the webpage's client-
side javascript file -->
<script src="/script.js" defer></
script>
```

Note: Glitch gives a very cool option to automatically format/beautify the file by doing some auto-aligning. You can select 'Format this file' from the GUI to enable this feature. Even a particular part of the code can be formatted with the same option, which changes to 'Format this selection' once a portion of the code is selected.

A Simple `<ul>` list names counts where our client side script can now populate the values returned by the server:

```
<main>
  <h2>The latest COVID-19 stats</
h2>

  <section class="counts">
    <ul id="counts">
      <em>loading counts&hellip;</
em>
```